

Ground-based Characterization of Small NEOs in the Context of NEOMIR

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September 15, 2026 15:40–16:00 — NEOMIR workshop: Hunting Asteroids near the Sun

Credit: The University of Tokyo

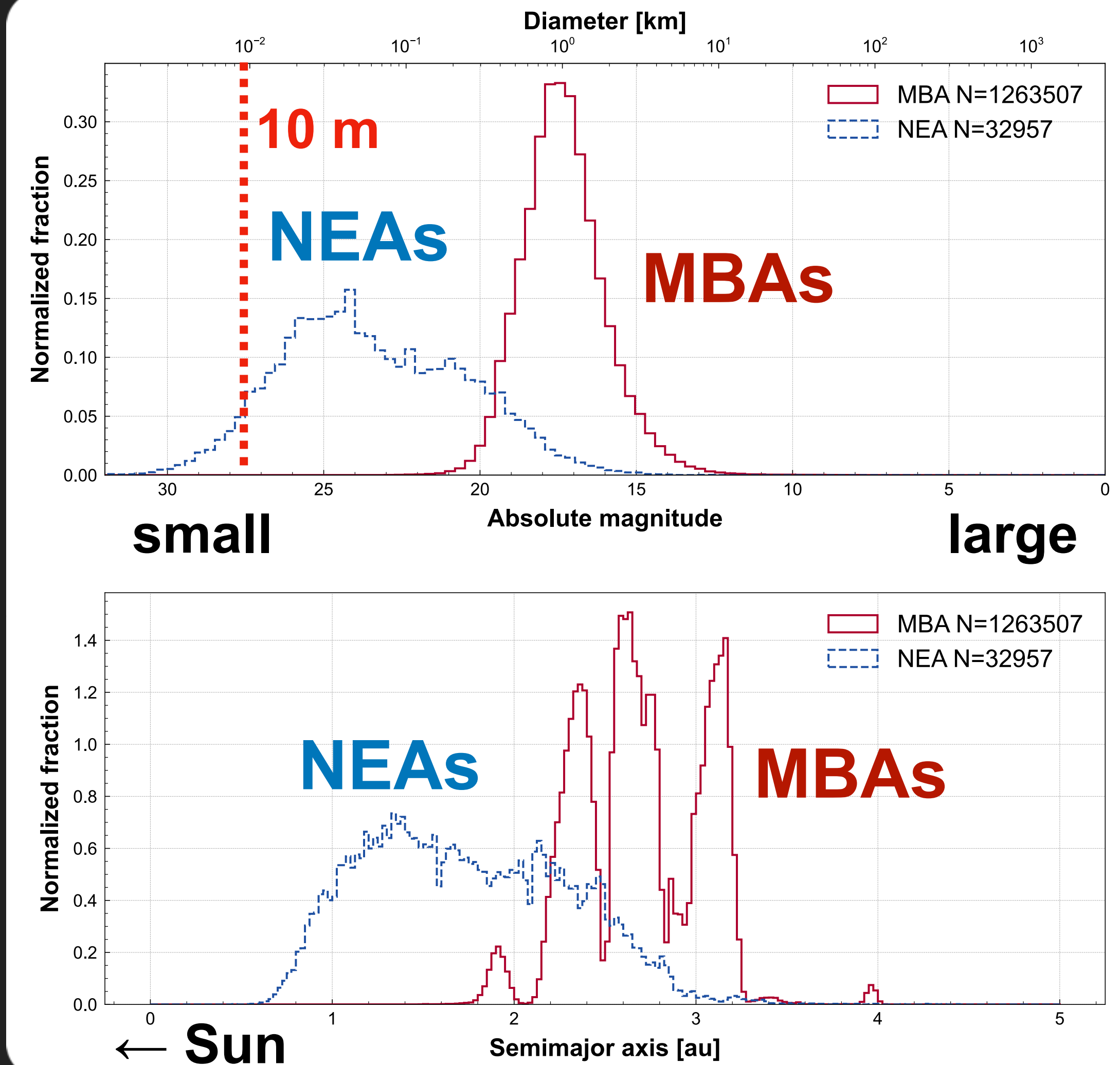
Small (< 100 m) Near-Earth Asteroids (NEAs)

- Unique population → Unique study

- ▶ Small → **Non-gravitational effects**
- ▶ Close to the Earth (Venus, Mars) → Orbit
- ▶ Up to $T \sim 1000$ K → Thermal effect

- Practically

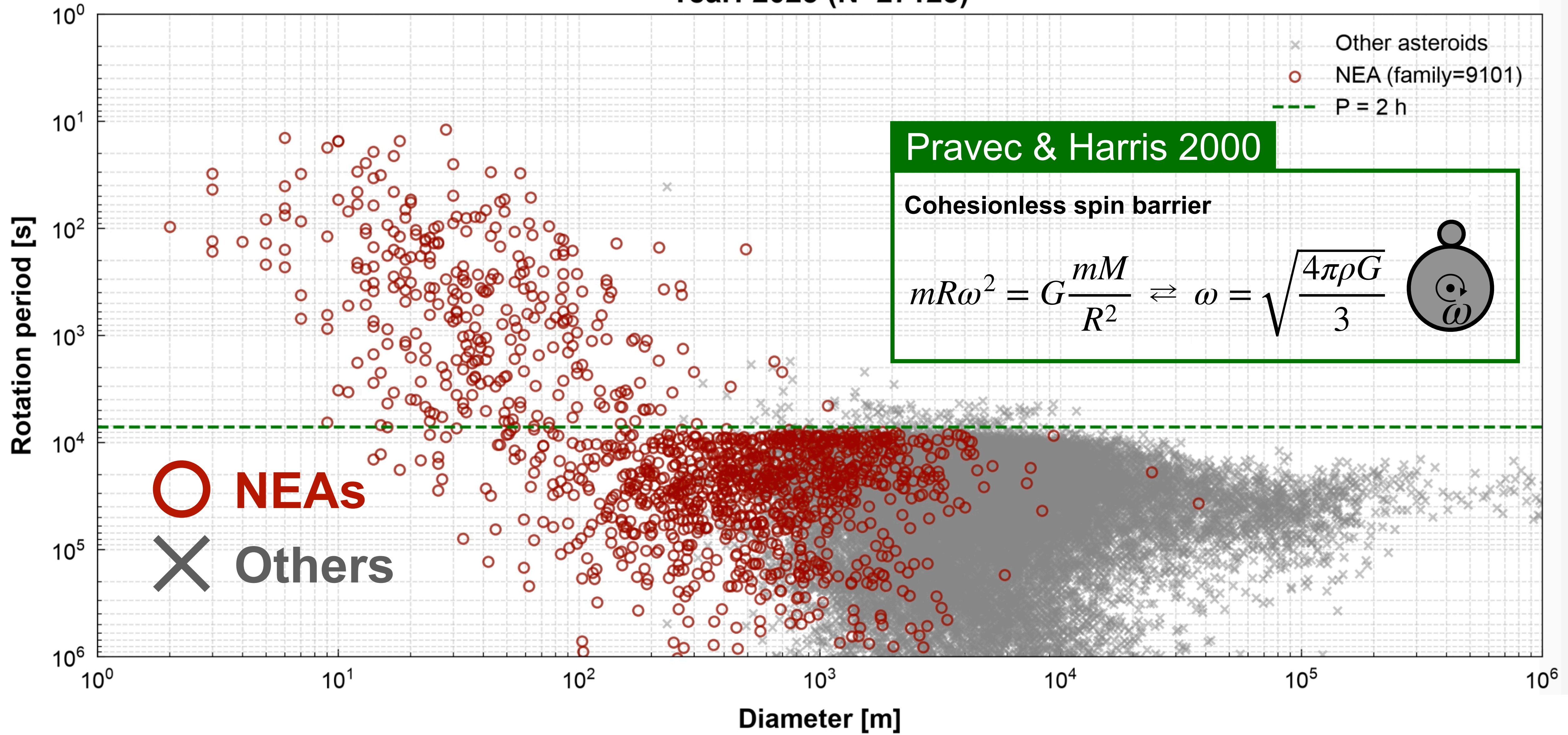
- ▶ Resources
- ▶ Planetary defense
 - Orbit (Discovery)
 - Diameter
 - Rotation
 - Composition
 - Spin pole, shape.....



Size (upper) and semimajor axes (lower) of MBAs and NEAs.

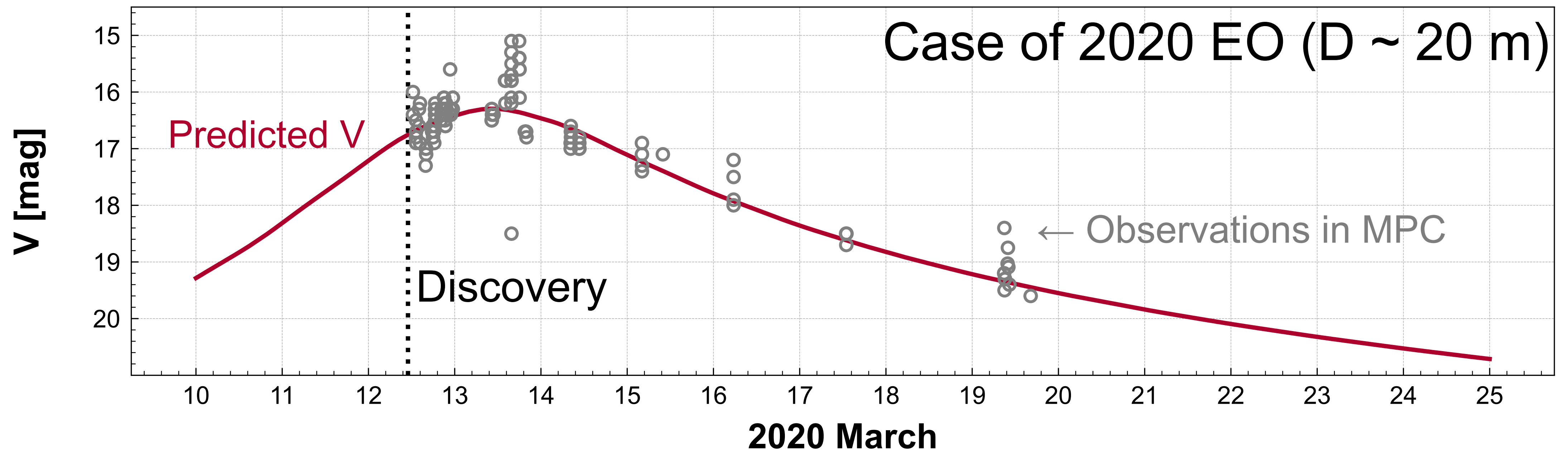
Asteroid Diameter vs. Rotation Period

Year: 2023 (N=27128)



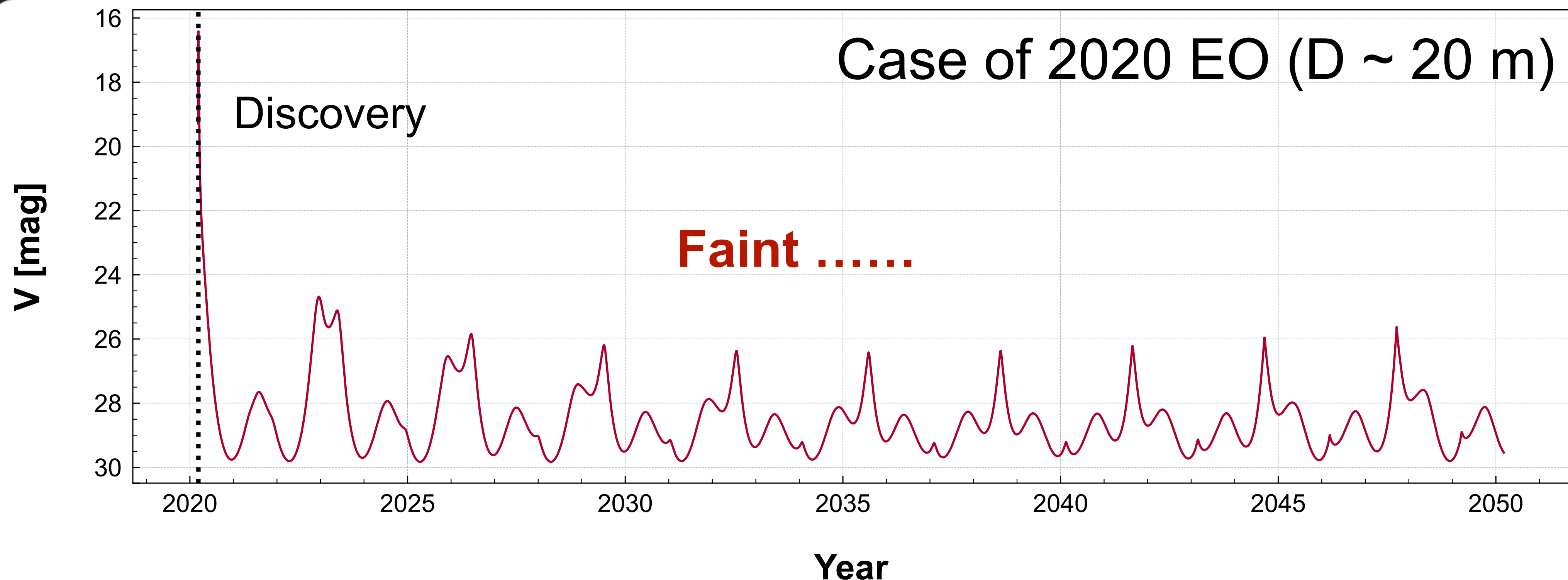
Challenges in Observing “Small” Asteroids

- Short observing window
- Large apparent motion
etc.....



Challenges in Observing “Small” Asteroids

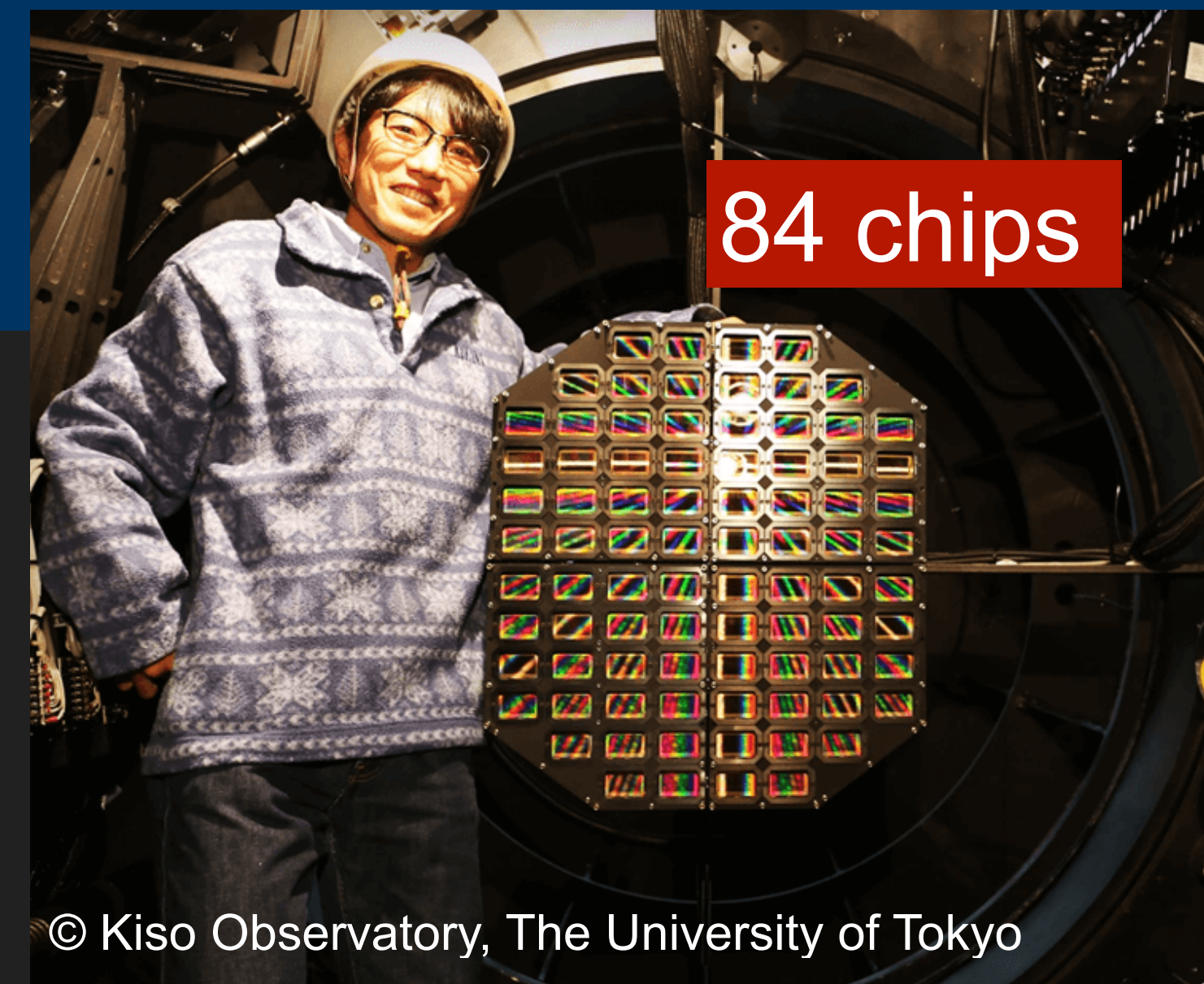
- Short observing window
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etc.....



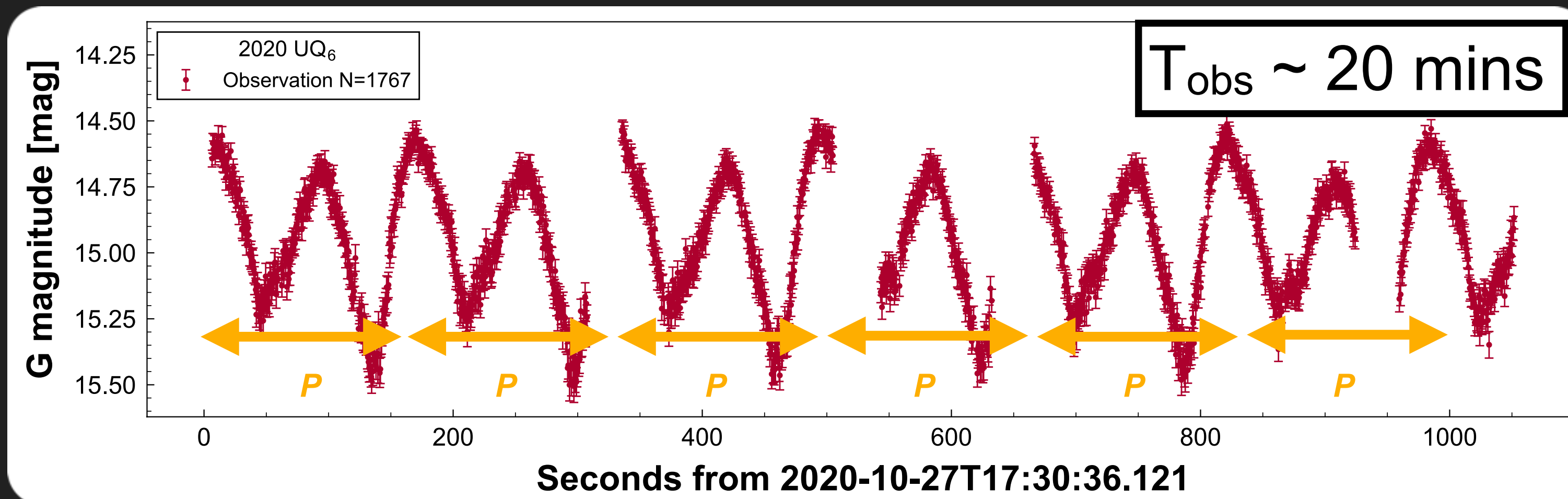
Lightcurve observations

- Instrument: 1.05 m Schmidt/Tomo-e Gozen
- Period : 2018.05 – 2023.11
- Targets : 108 small NEAs
Nominal criteria $H > 22.5$ & $V < 17$ mag
($\Leftrightarrow D < 100$ m)

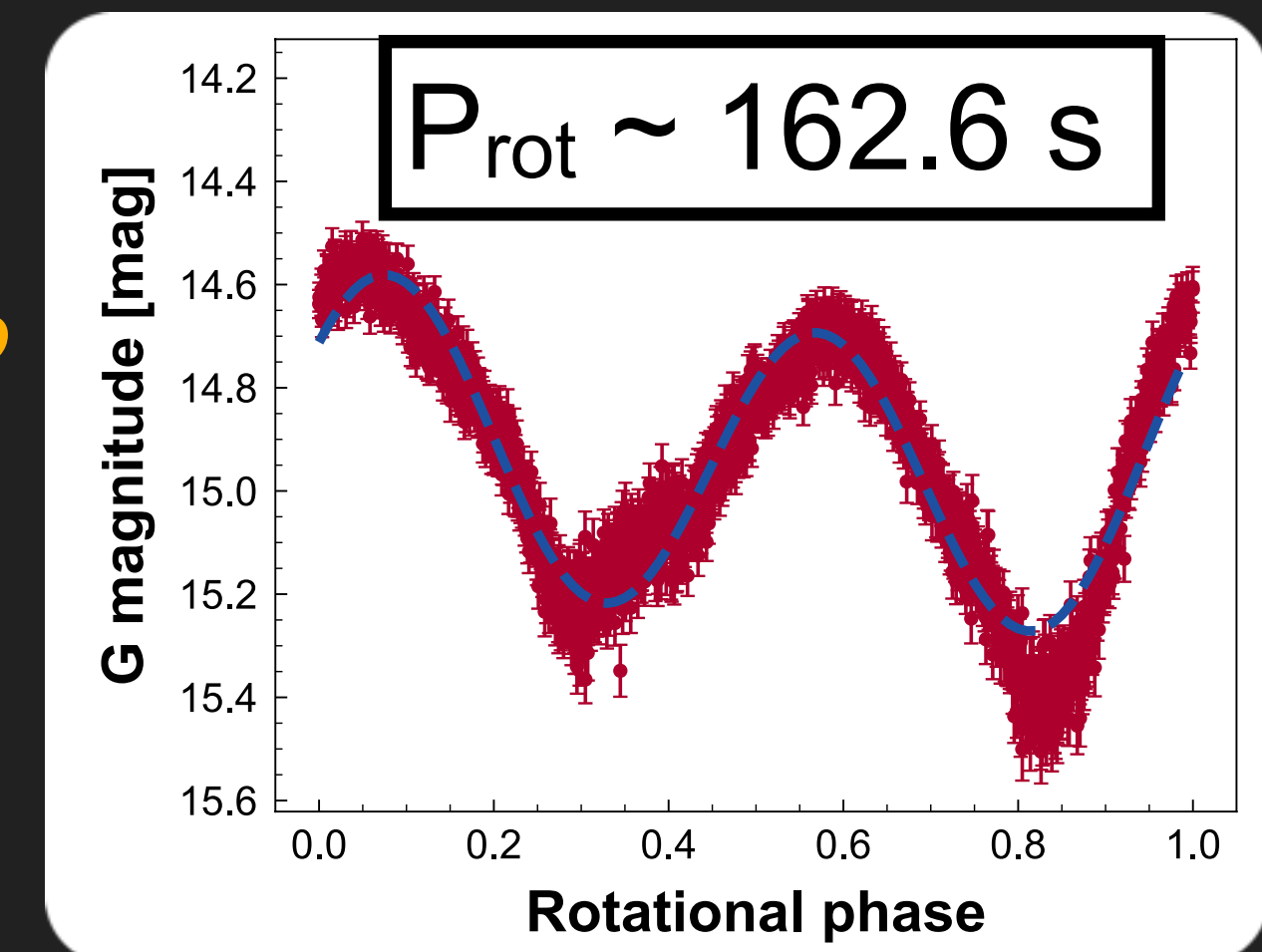
36/108 discovered by
Tomo-e Gozen



Tomo-e Gozen camera

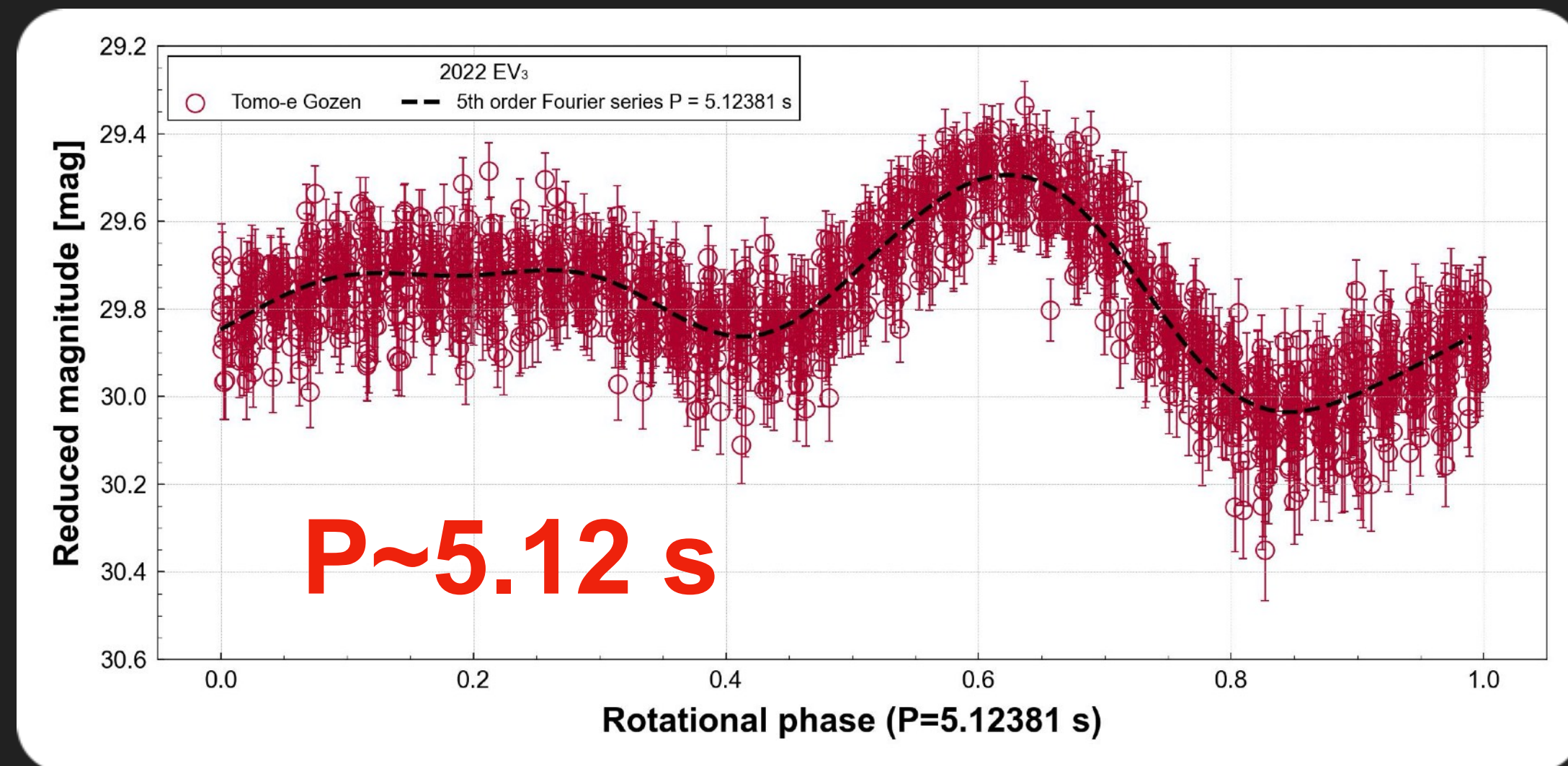


Fold with P

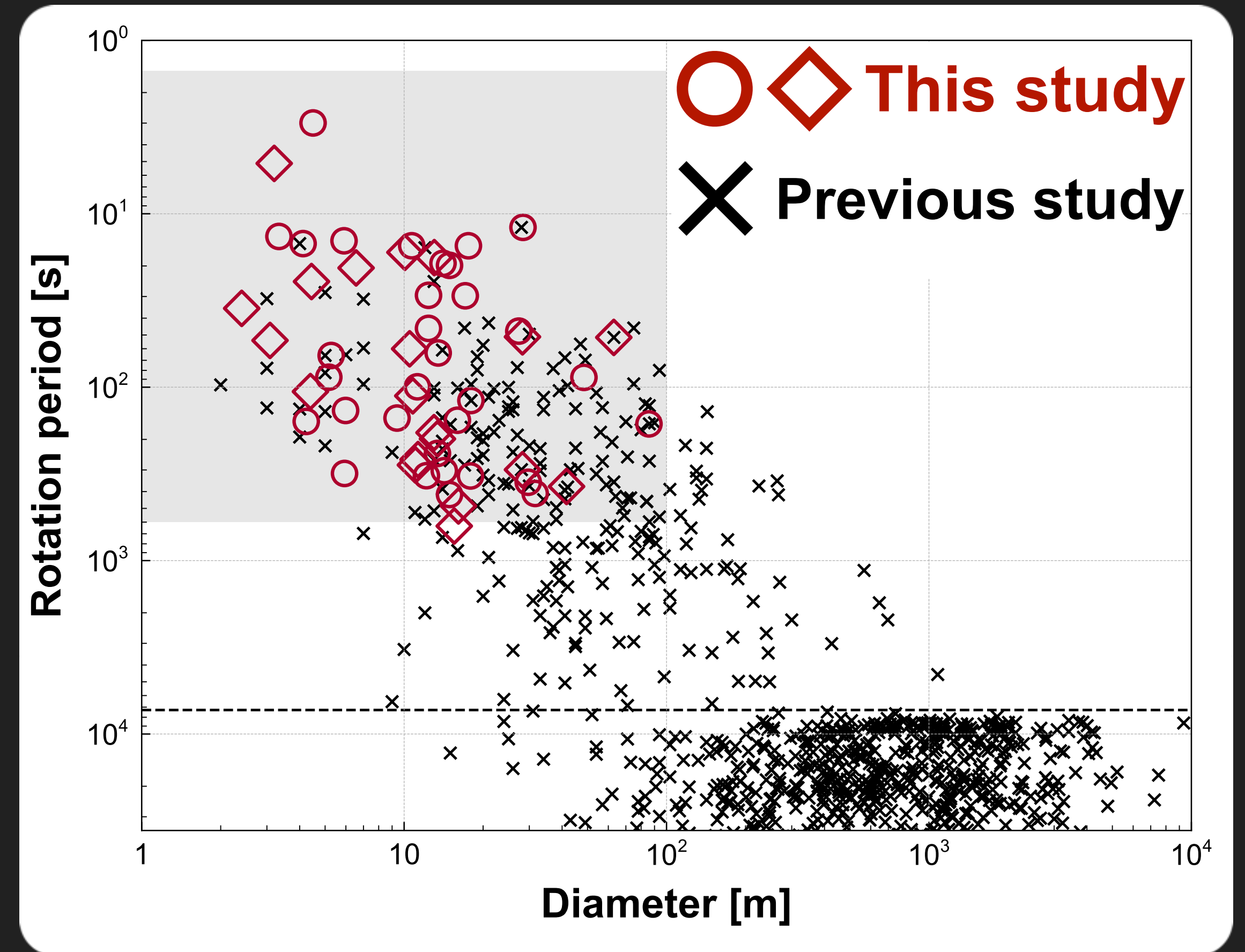


Result. Diameter-rotation period relation

Rotation periods of 52 (out of 108) NEOs were derived.

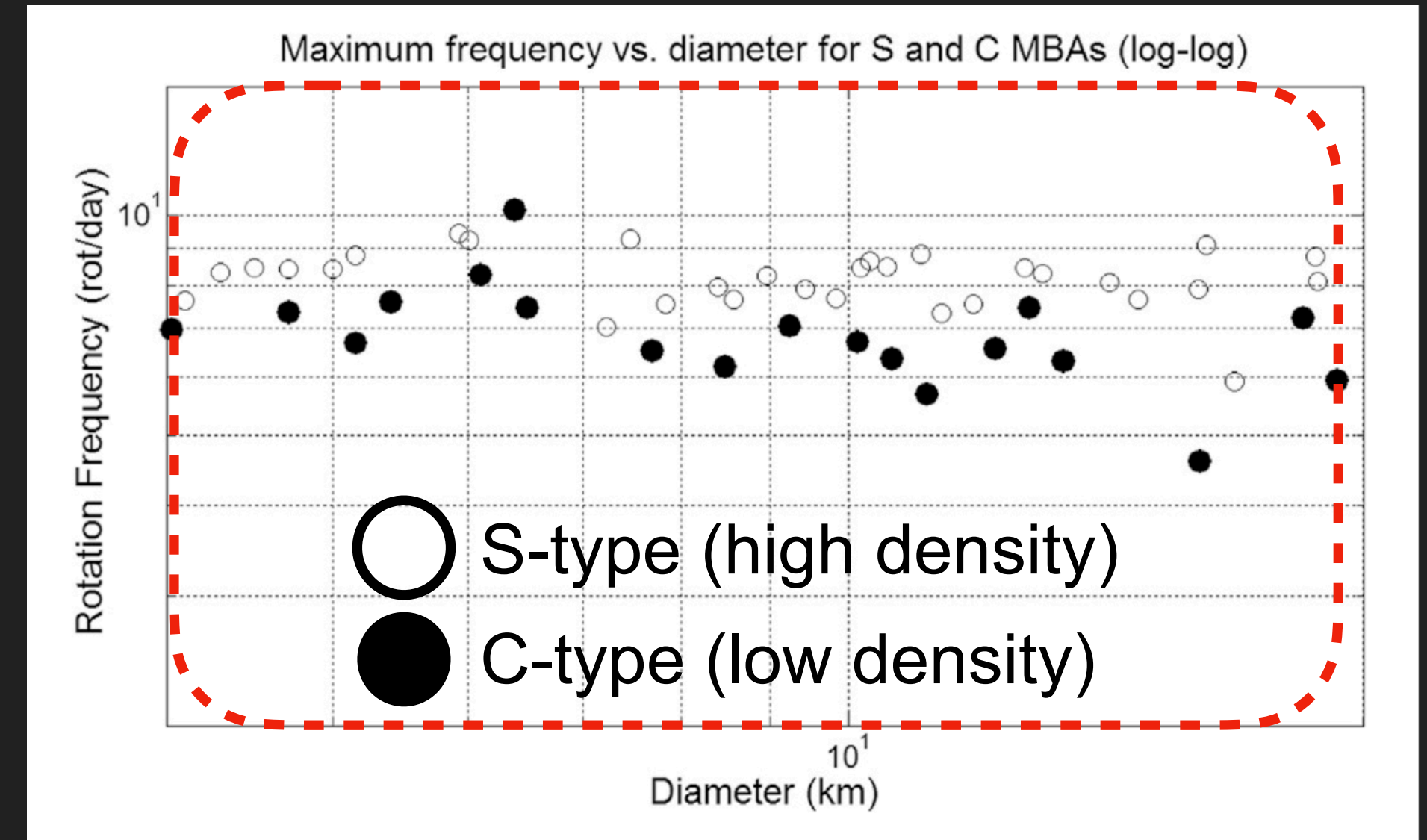
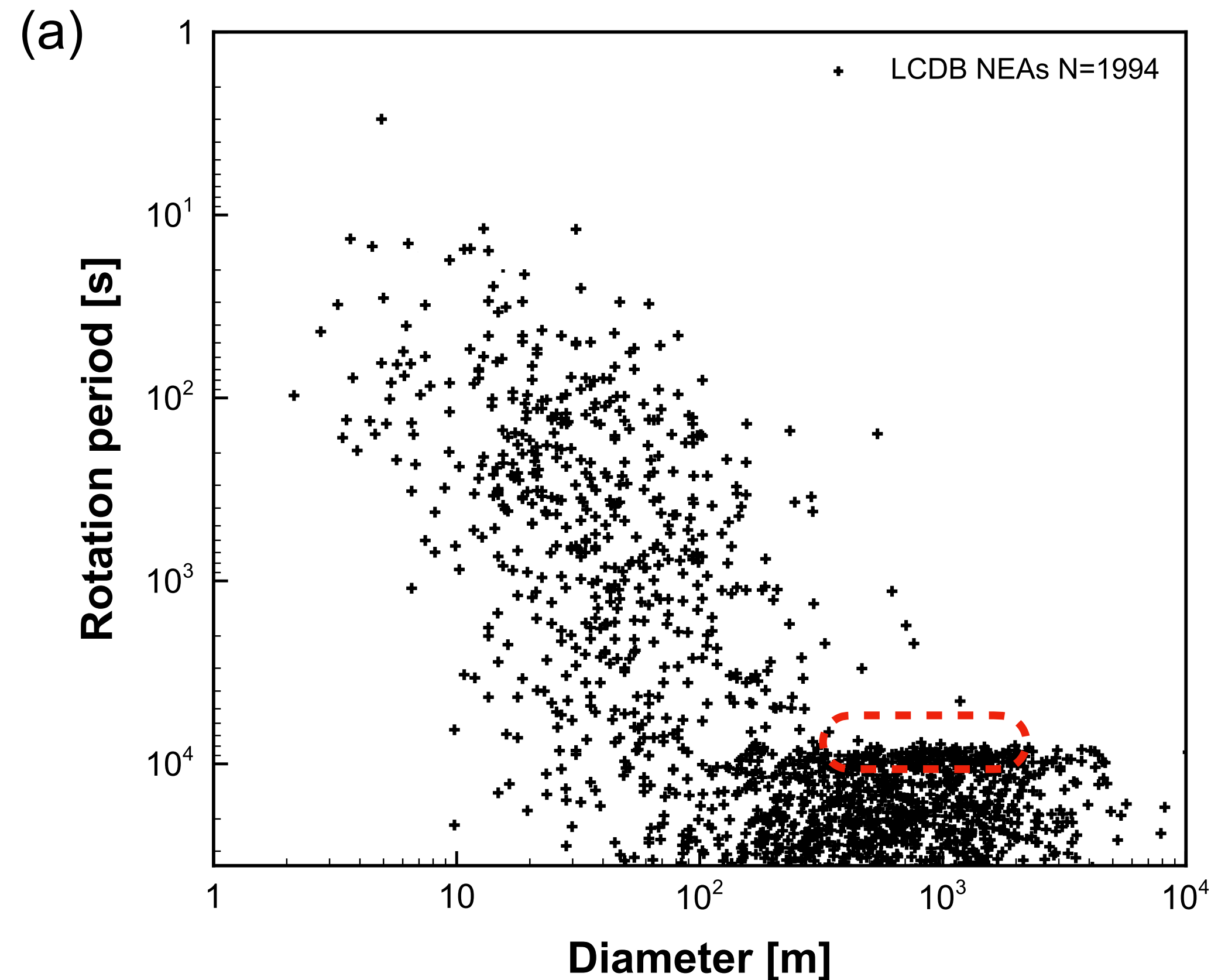


Lightcurves of 2022 EV₃ (H=29.84)



Diameter-rotational period relation

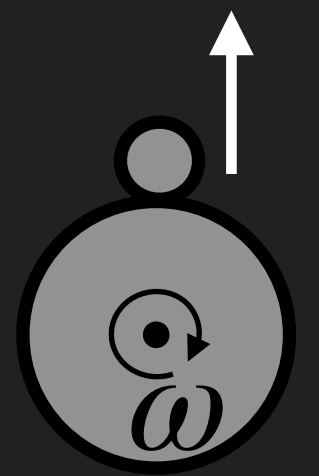
Diameter vs. Rotation periods



Pravec & Harris 2000

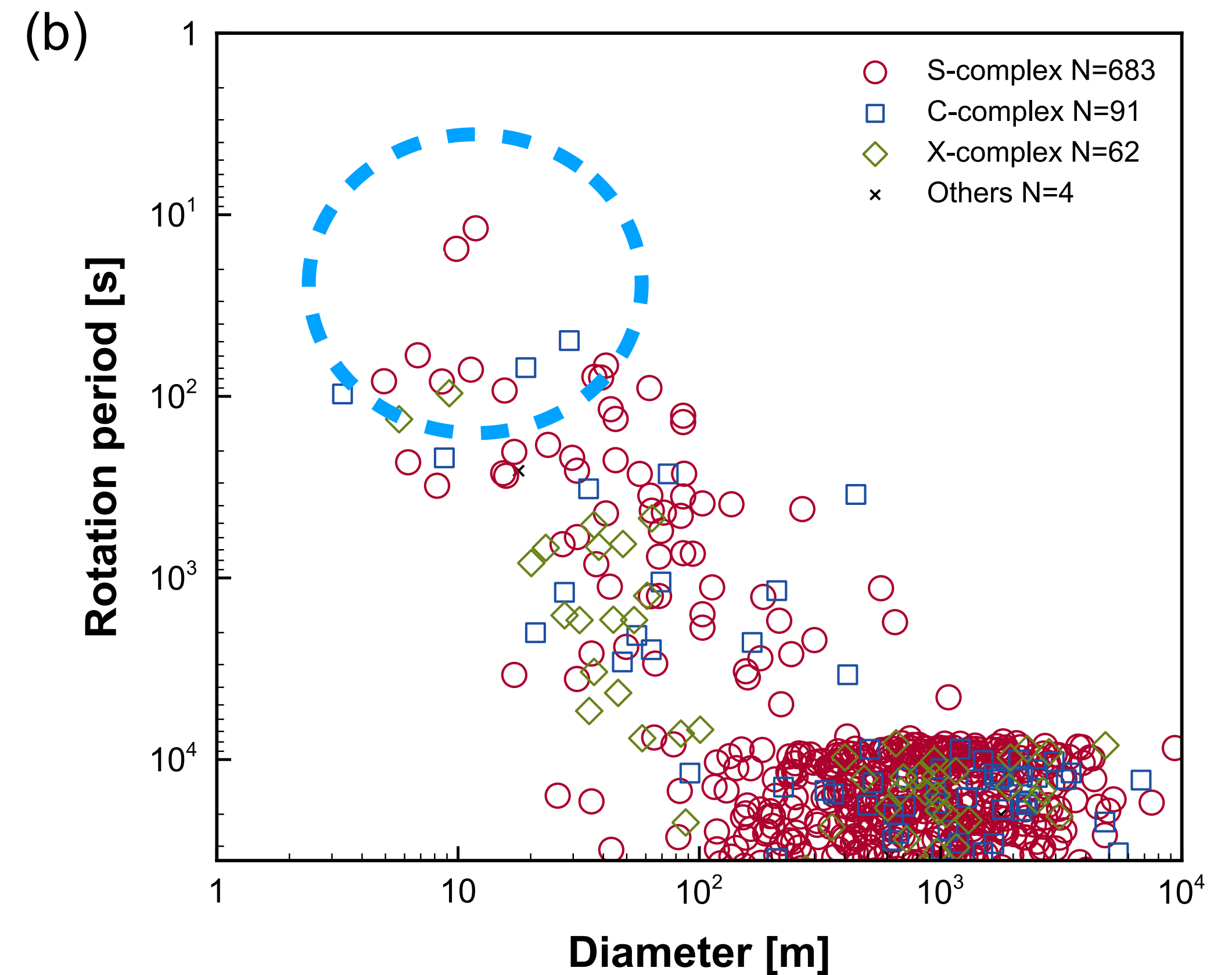
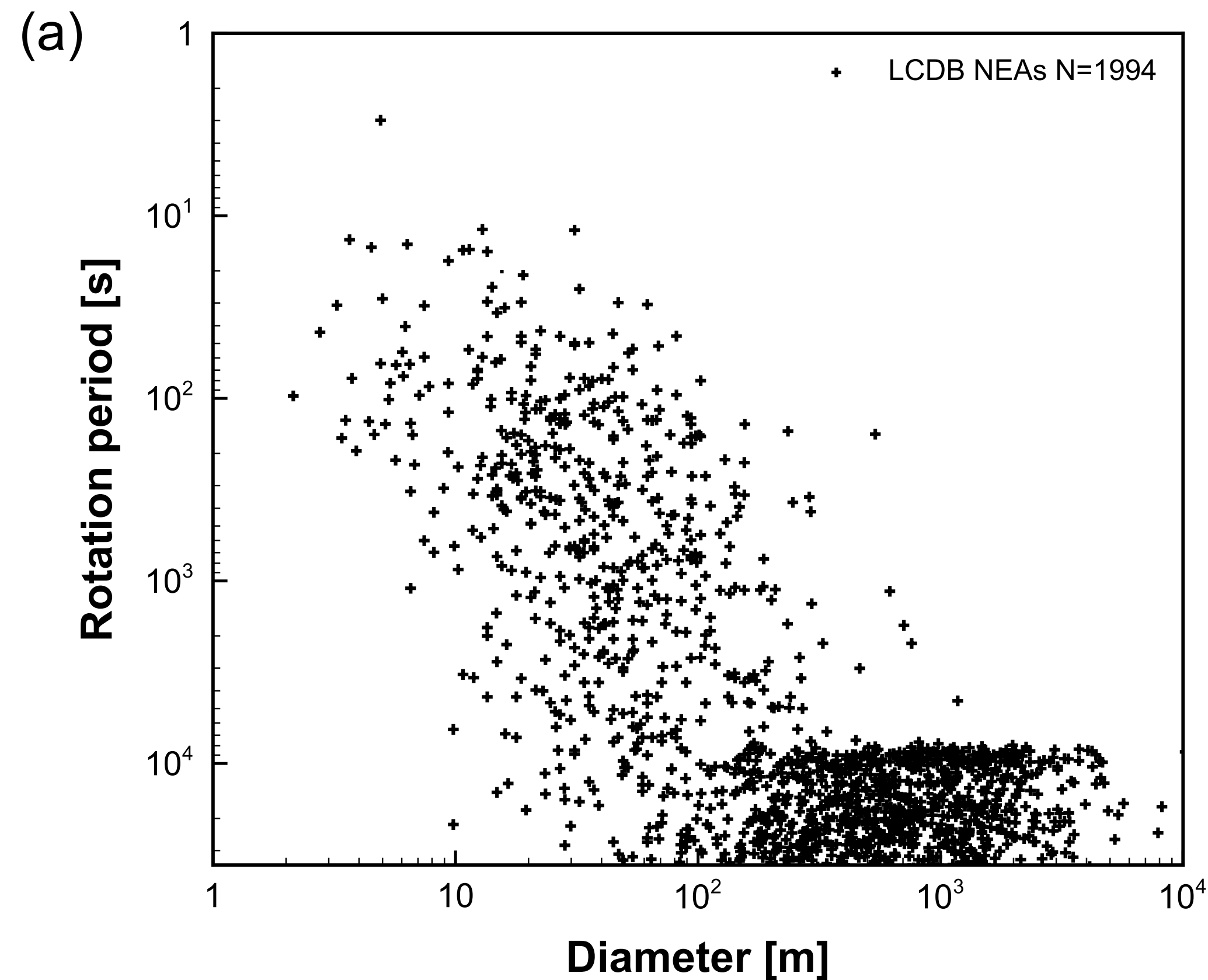
Cohesionless spin barrier

$$mR\omega^2 = G \frac{mM}{R^2} \Leftrightarrow \omega = \sqrt{\frac{4\pi\rho G}{3}}$$



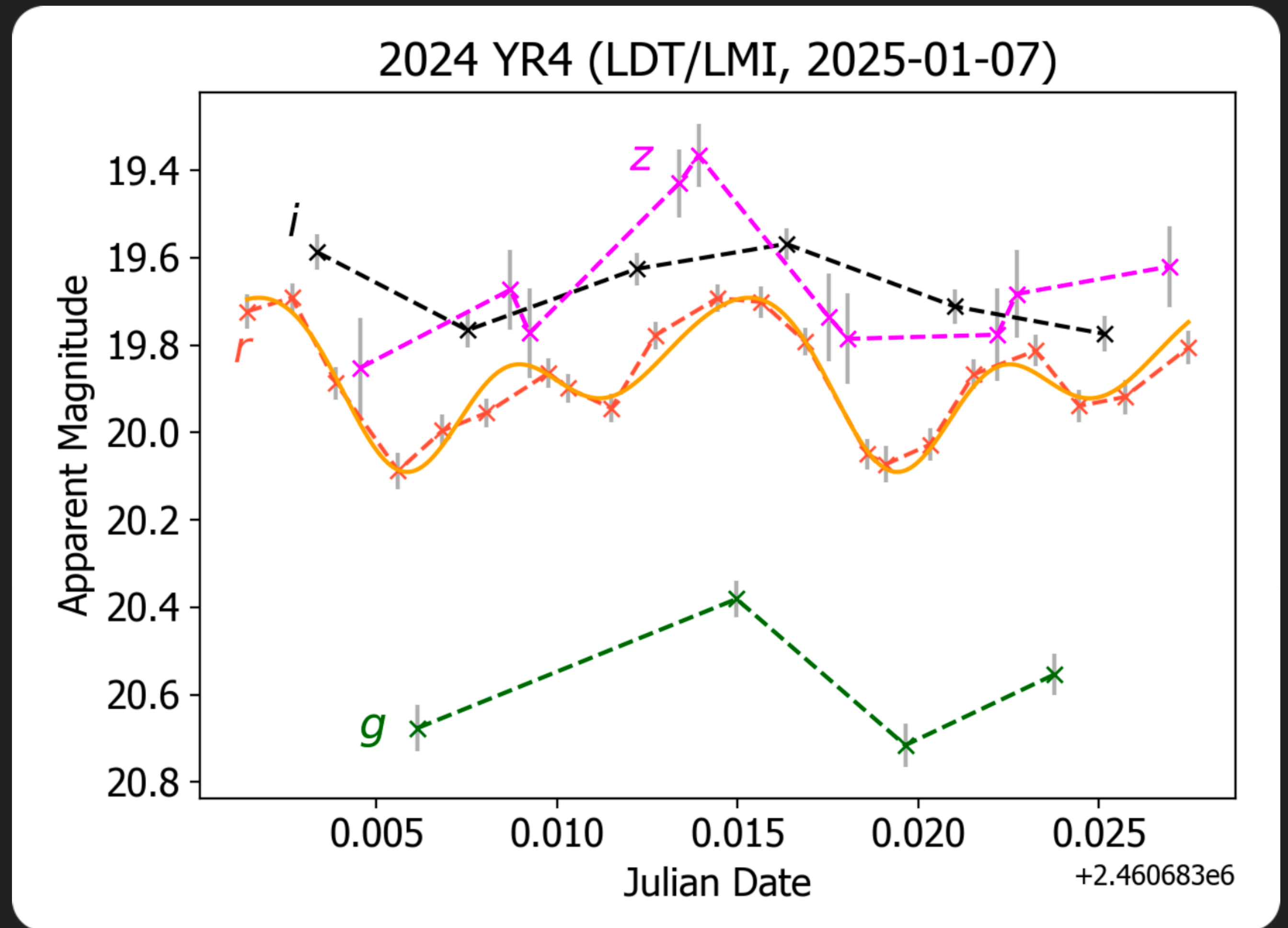
Diameter vs. Rotation periods

- Small fast rotators remain poorly studied.



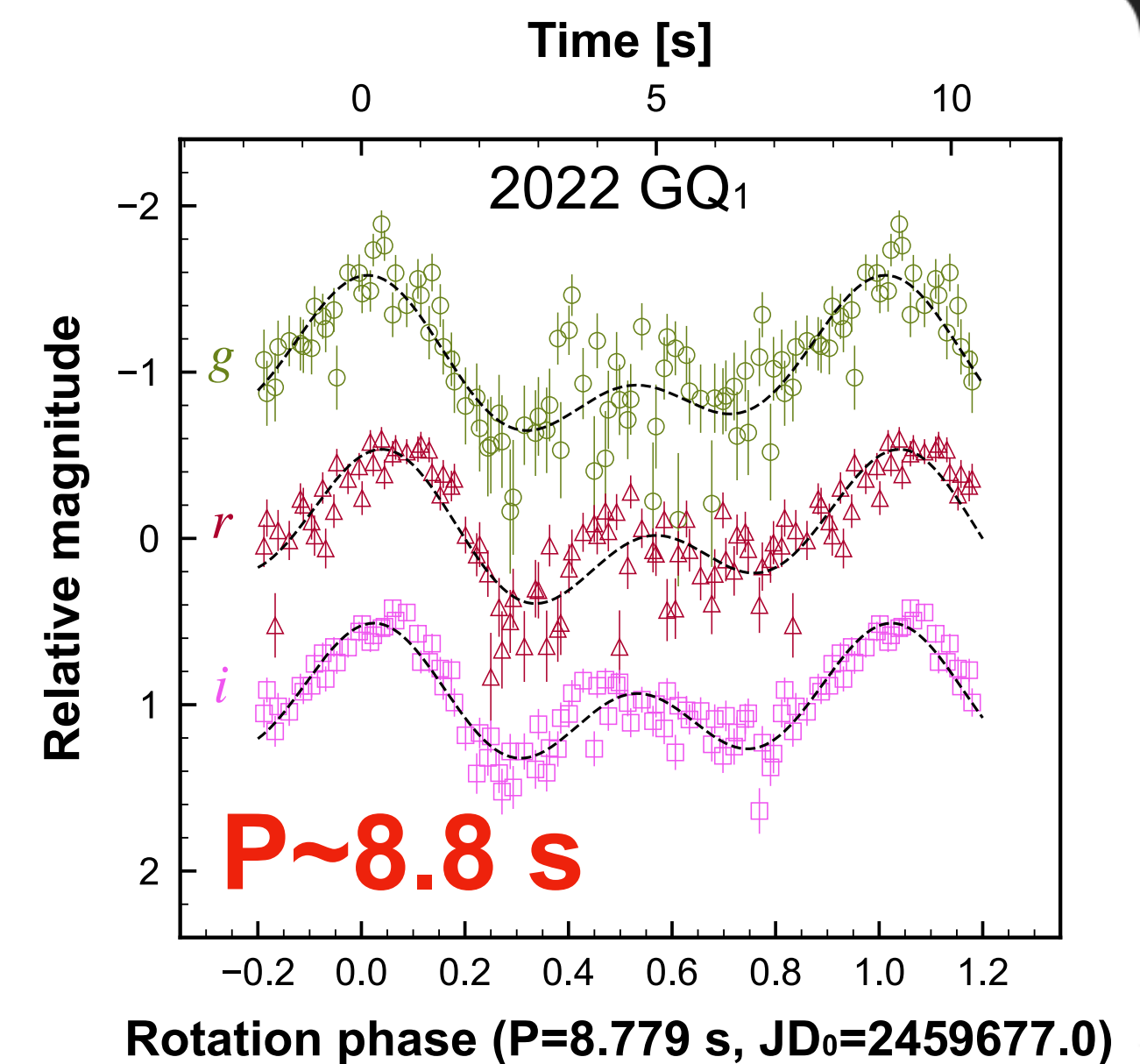
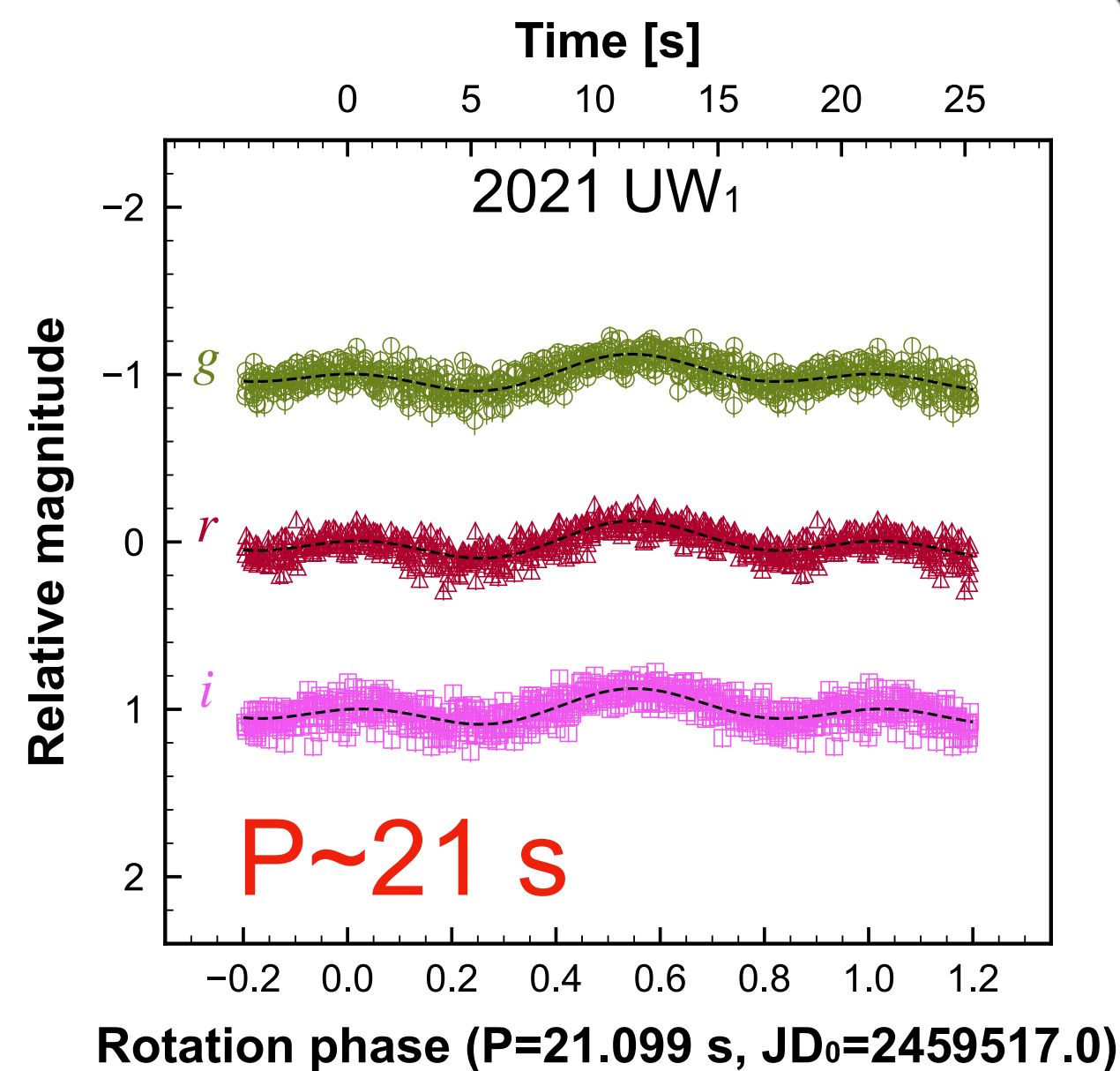
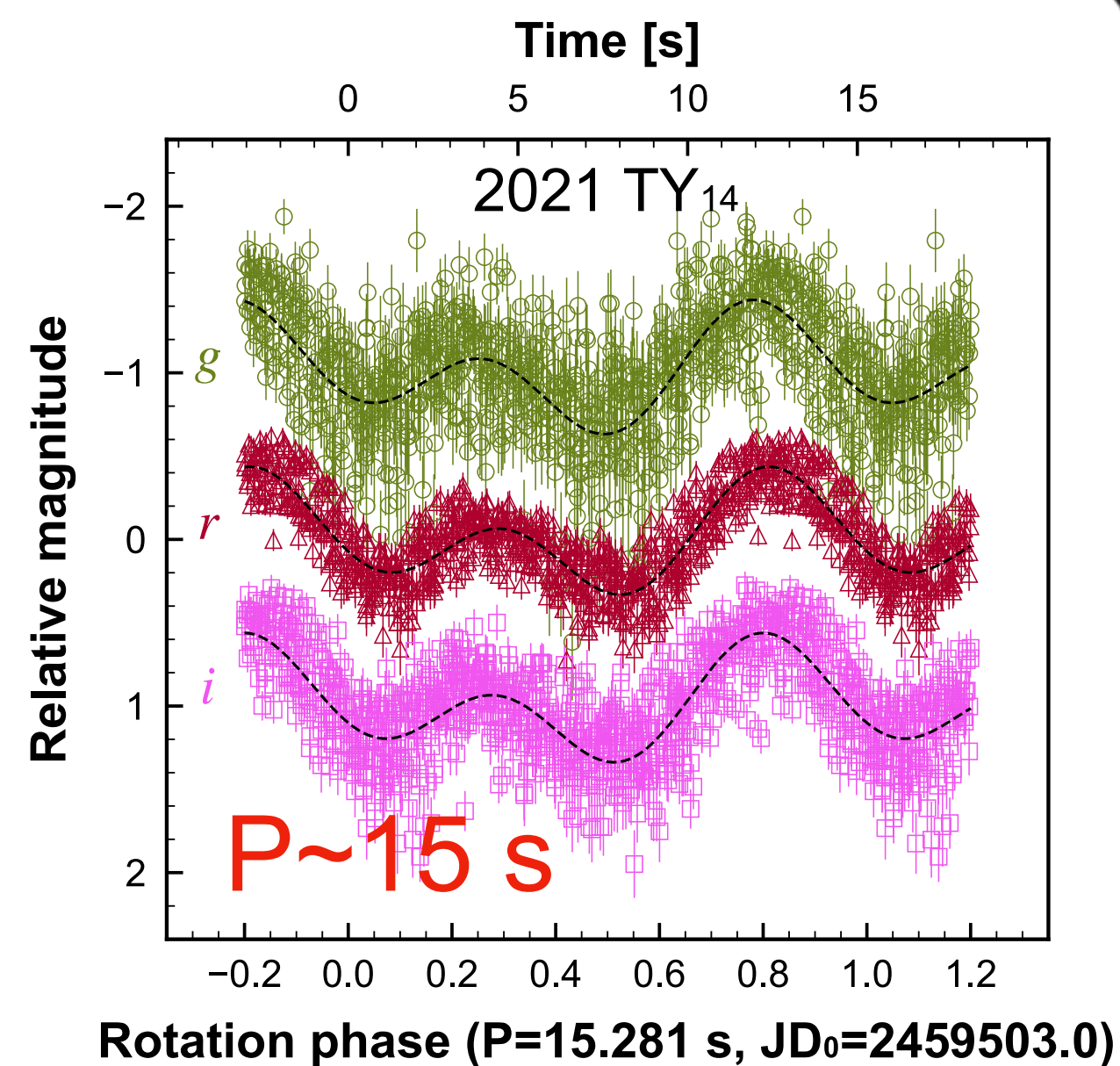
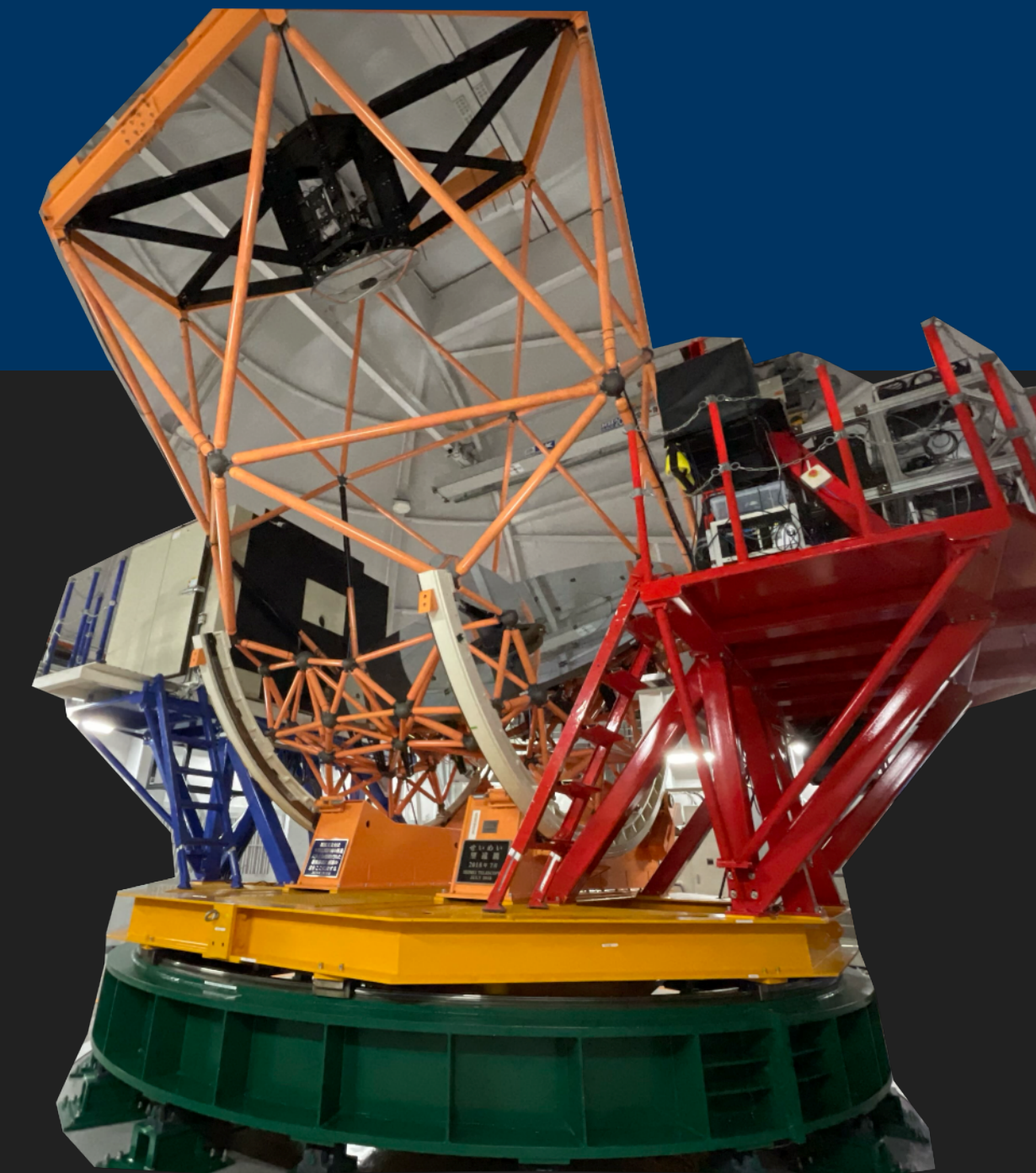
Difficulty in Color Estimate of Fast Rotators

Sequential multi-filter observations are affected by lightcurve variations



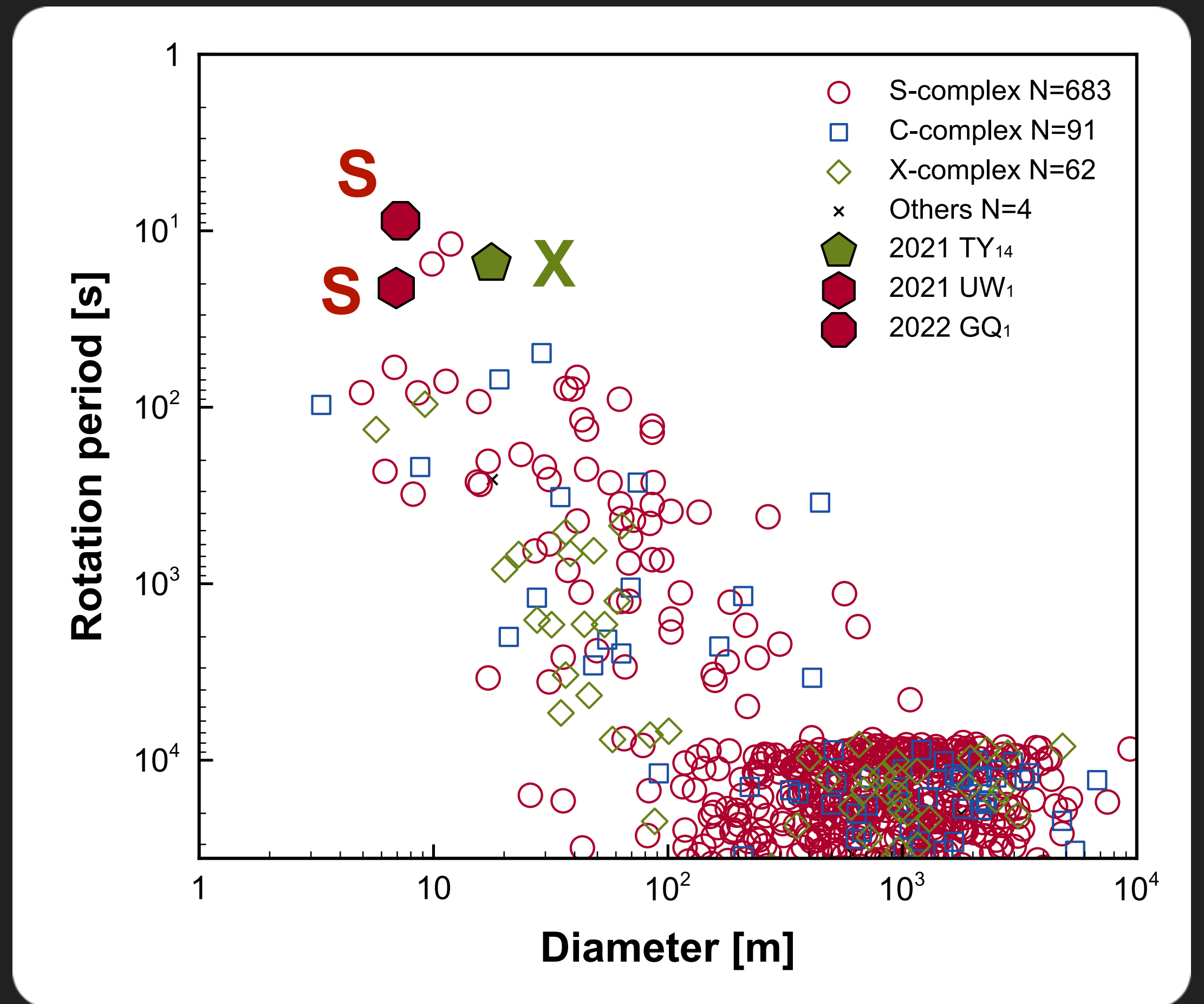
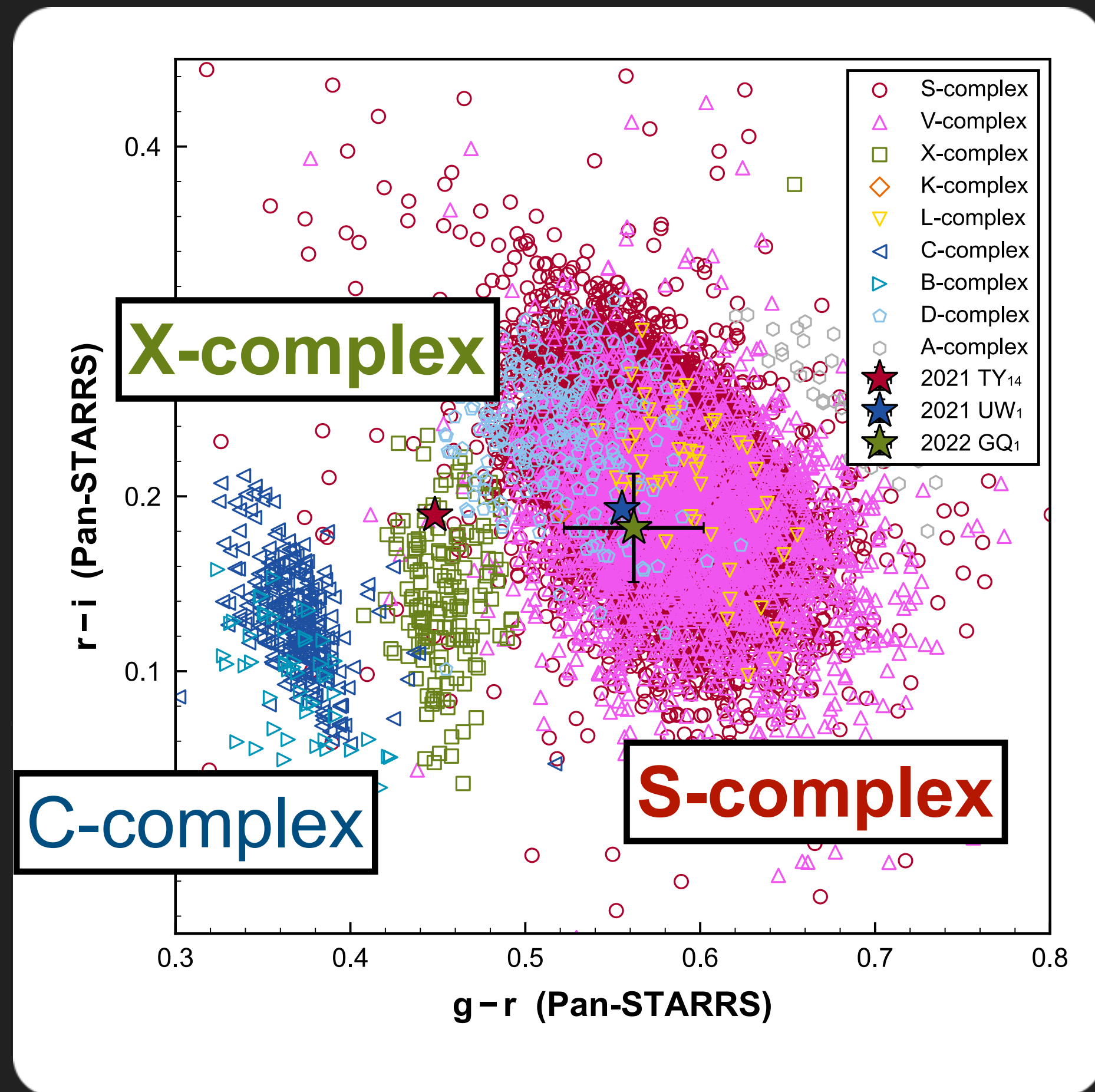
Seimei/TriCCS observations

- Instrument: 3.8 m Seimei telescope / TriCCS in Japan
 - 1 s / 5 s exposures
 - *g*, *r*, and *i* bands
- Period: 2021 and 2022
- Targets: 3 tiny asteroids with sub-minute rotations



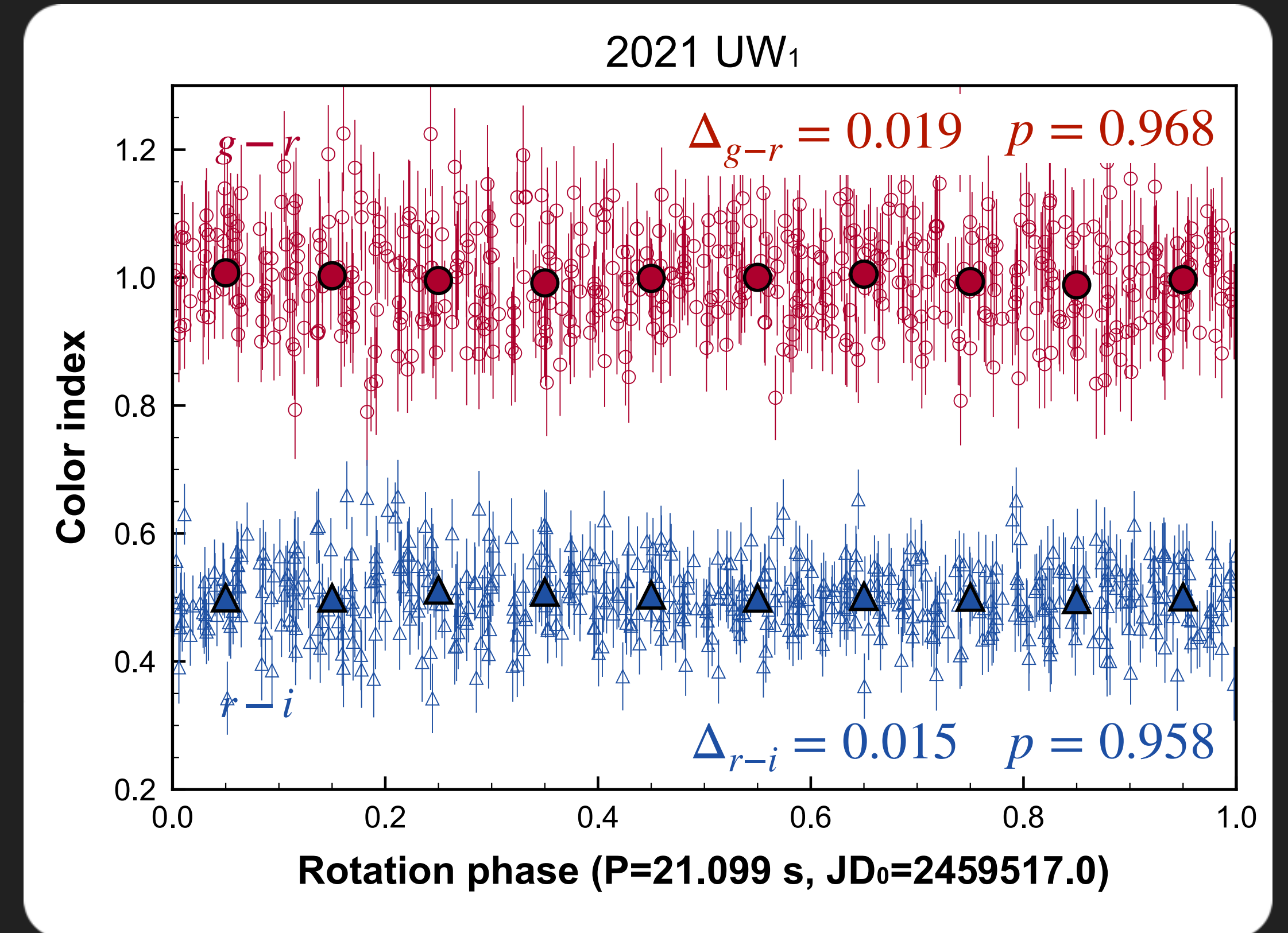
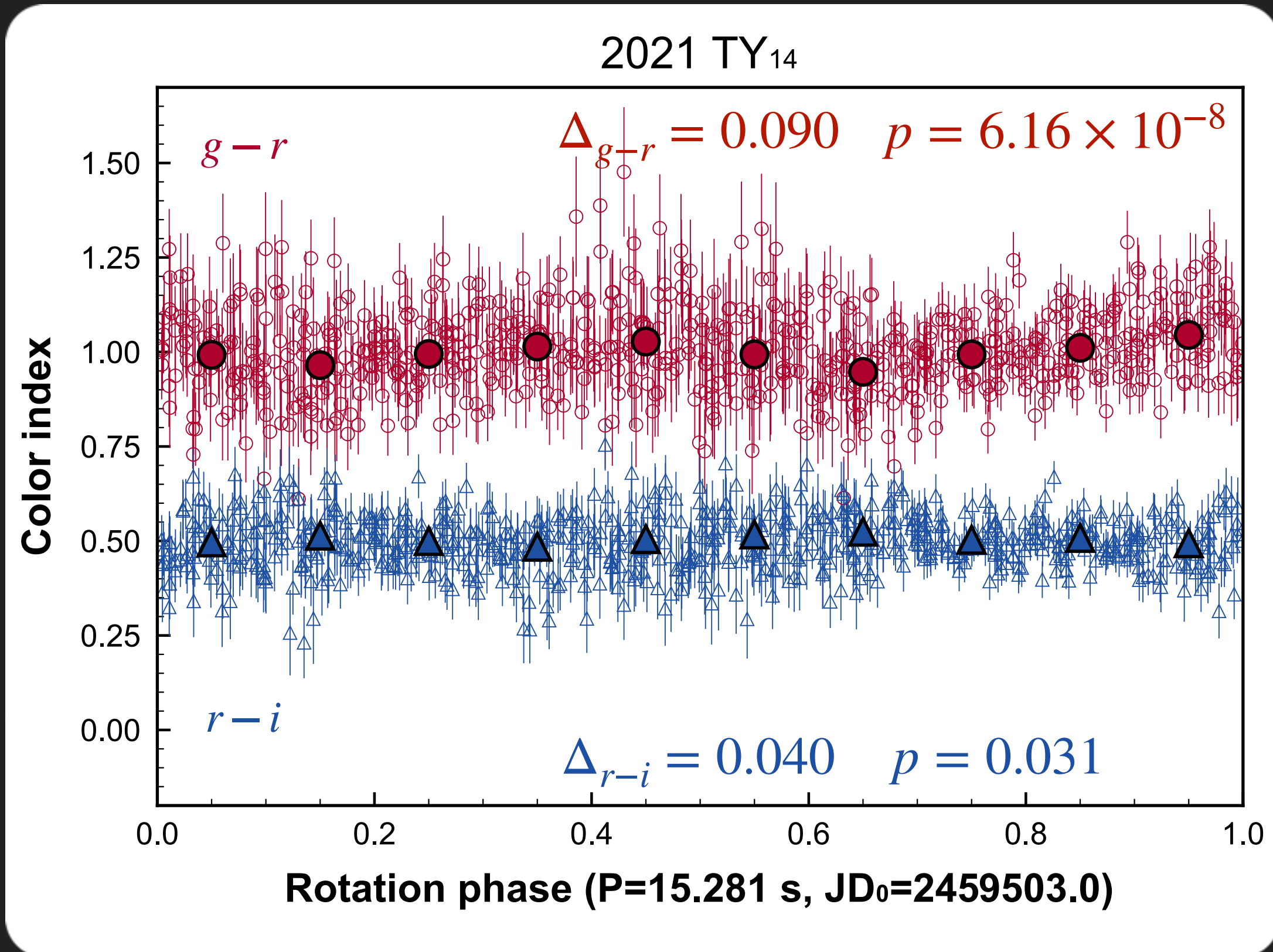
Colors and Spectral Types

- Fastest rotator ever characterized by spectrophotometry



Heterogeneity

- Weighted one-way analysis of variance (ANOVA)
 - Null hypothesis: constant color across the rotation phase



Future prospects

- Rubin & NEOSurveyor → lots of discovery and characterizations
- **TAO**: the University of **T**okyo **A**tacama **O**bservatory
 - ▶ Co. Chajnantor in the Atacama Desert (5640 m)
precipitable water vapor (PWV) ~ 0.5 mm
 - ▶ 1 m telescope (miniTAO, 2009–, Müller+2013, 2005 YU55)
6.5 m telescope (soon)
 - ▶ Instruments
 - MIMIZUKU (2–38 μm)
 - Laguna (visible, CMOS)



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Co. Chajnantor 5640m



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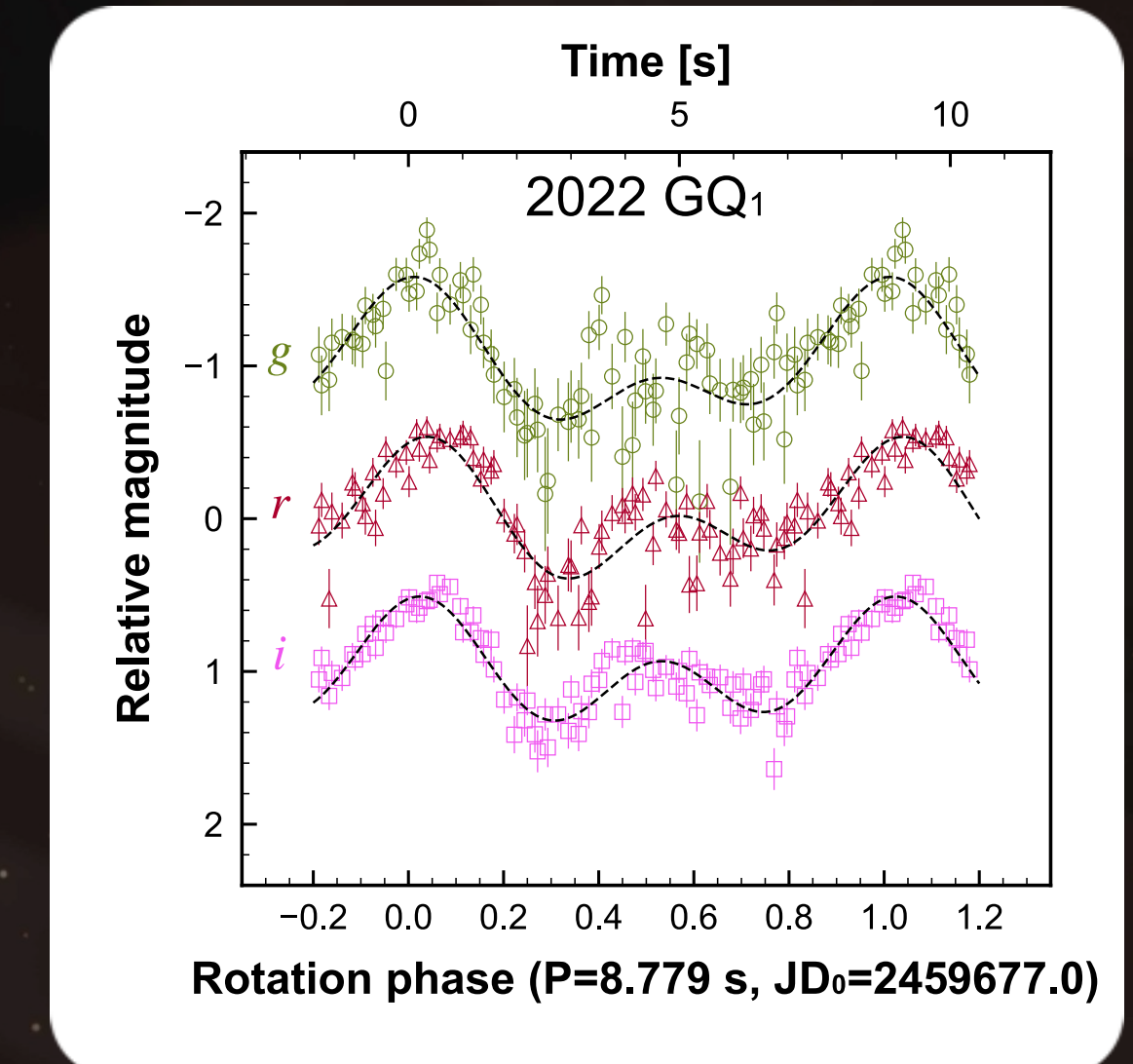
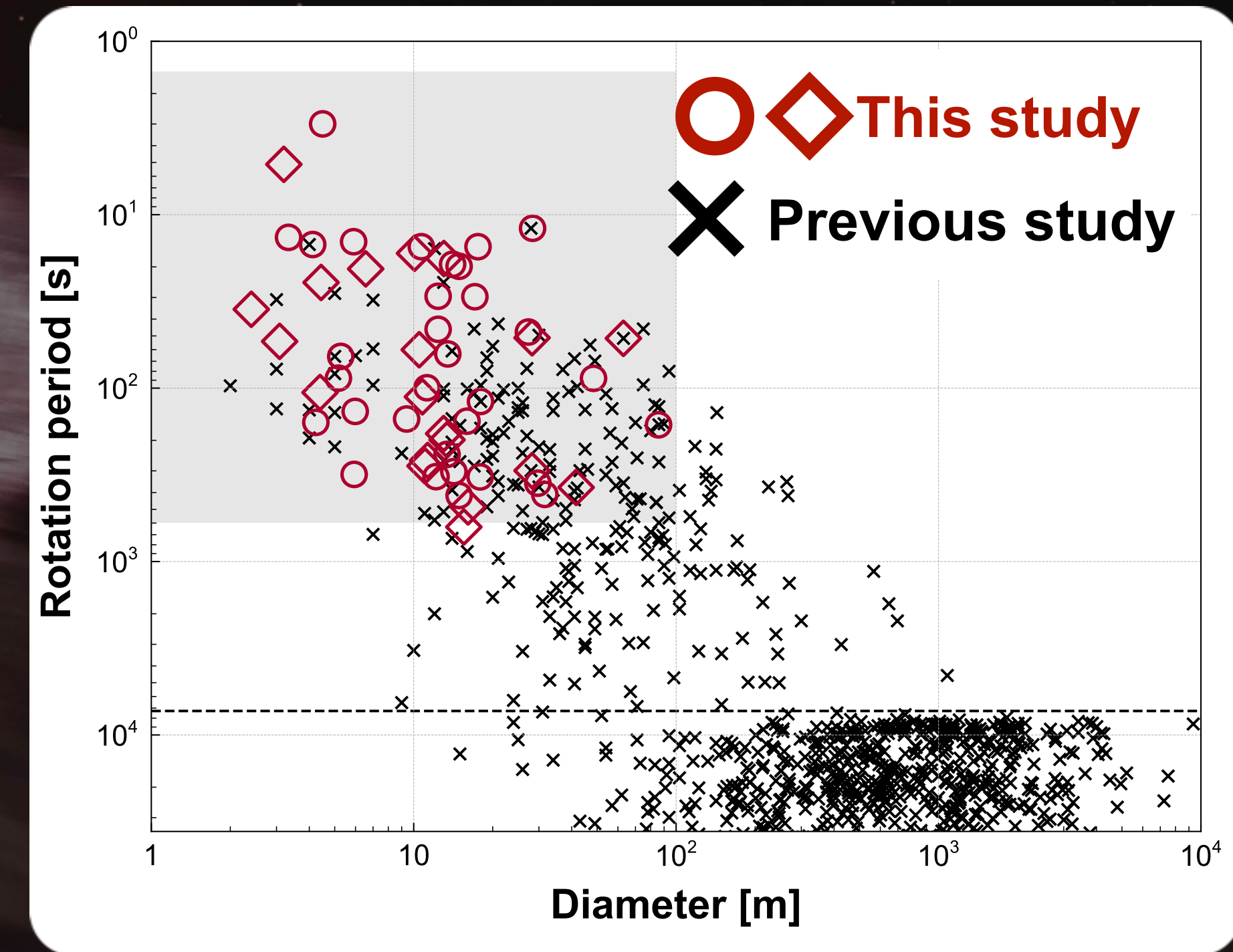
TAO

https://www.u-tokyo.ac.jp/focus/en/press/z0508_00344.html
<https://www.ioa.s.u-tokyo.ac.jp/TAO/news/20240430/index.html>
<https://www.eso.org/public/images/potw1150a/>
https://www.nayoro-obs.jp/binarystar2017/slides/S3-1_Takahashi.pdf

Müller et al., 2013, A&A, Vol. 558, A97.
Kamizuka et al., 2022, Proc. of SPIE. Sako et al., 2026, Proc. of SPIE.

Takeaways

- Observations of NEAs at close approach provide exciting scientific opportunities.
- Tiny NEAs are rotating as fast as seconds. Fast (video) observations are critical to characterize them.
 - Rotation periods
 - Surface composition & heterogeneity



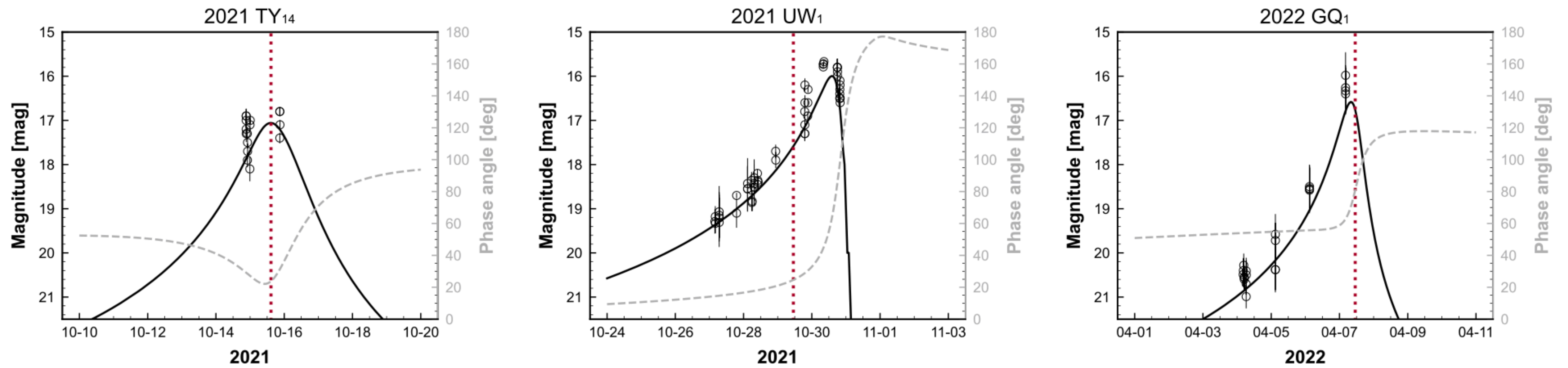


Figure 1. Ephemerides of 2021 TY₁₄, 2021 UW₁, and 2022 GQ₁. Apparent brightnesses and phase angles are shown by solid and dashed lines, respectively. The times of our observations are indicated by vertical dotted lines. Photometric data from the Minor Planet Center (MPC) database (https://www.minorplanetcenter.net/db_search, accessed on 2025 October 31) with reported magnitude uncertainties are shown as circles with the error bars. For illustrative purposes, all available photometric data are plotted regardless of the calibrated band.

Table 1
Summary of the Observations

Object	Obs. Date (UTC)	t_{exp} (s)	N_{frame}	V (mag)	α (deg)	v (arcsec s ⁻¹)	Air Mass	Seeing (arcsec)
2021 TY ₁₄	2021 Oct 15 14:42:17–14:59:02	1	752	17.0	23.7	2.10	1.00–1.00	3.2
2021 UW ₁	2021 Oct 29 11:02:22–11:52:41	5	525	17.6	24.5	0.42	1.10–1.20	3.5
2022 GQ ₁	2022 Apr 07 11:04:31–11:05:50	1	80	16.7	81.3	7.35	1.62–1.62	2.5

Note. Observation time in UT in midtime of exposure (Obs. Date), exposure time per frame (t_{exp}), and the number of frames (N_{frame}) are listed. Predicted V band apparent magnitude (V), phase angle (α), and apparent angular rate of asteroids (v) at the observation starting time refer to NASA Jet Propulsion Laboratory (JPL) Horizons as of 2026 May 10. Elevations of asteroids to calculate air mass range (Air Mass) also refer to NASA JPL Horizons. Seeing FWHMs (Seeing) in the r band for Seimei observations measured by computing the FWHMs of reference stars are also listed.